

COMPILE TEST FOR THE F2 FRAGMENT

Anonymous authors

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This wrapper exists only to compile `results/F2_R6_validity.tex` under the conference template. The citation (Hinton et al., 2006) exercises bibtex.

1 VALIDITY OF THE REDUCED LP INEQUALITIES

This section verifies that each of the four constraint families of the reduced $O(K^2)$ linear program is a valid inequality: for every instance and every run of predictive greedy on it, the quantities read off the run satisfy the constraint. Consequently the LP is a relaxation of the set of realisable runs, its optimum is a lower bound on the worst-case ratio for every K , and the closed form of that optimum inherits the same status.

1.1 SETTING AND NOTATION

Let N be a finite ground set and let $f: 2^N \rightarrow \mathbb{R}$ be monotone and submodular with $f(\emptyset) = 0$. Write $d_e(S) = f(S \cup \{e\}) - f(S)$ for the marginal gain of $e \notin S$. The predictor $\tilde{f}: 2^N \rightarrow \mathbb{R}$ is an arbitrary set function with $\tilde{f}(\emptyset) = 0$; it is not assumed monotone or submodular. Its gains are written $\tilde{d}_e(S)$.

Definition 1 (error band). *For $\eta_u, \eta_o \geq 1$ we say that $\tilde{f}$ is (η_u, η_o) -accurate if for every $S \subseteq N$ and every $e \notin S$*

$$\frac{1}{\eta_u} d_e(S) \leq \tilde{d}_e(S) \leq \eta_o d_e(S). \quad (1)$$

We call the right inequality of equation 1 the upper band and the left inequality the lower band, following the constraint names `band_up` and `band_lo` in the LP builders. The global error is $\eta = \eta_u \eta_o \geq 1$. Gains are assumed strictly positive on the states that occur below, which is the regime in which equation 1 has content.

Predictive greedy starts from $S^0 = \emptyset$ and for $t = 0, \dots, K-1$ picks

$$b_t \in \arg \max_{e \notin S^t} \tilde{d}_e(S^t), \quad S^{t+1} = S^t \cup \{b_t\}, \quad (2)$$

and returns S^K . Ties in equation 2 are broken adversarially. We refer to equation 2 as the greedy choice.

Fix an optimal set O with $|O| \leq K$ and $f(O) = \max_{|T| \leq K} f(T)$. Enumerate $O = \{o_1, \dots, o_K\}$ and normalise $f(O) = 1$. If $|O| < K$, set $g_{t,i} = 0$ for the $K - |O|$ unused indices i ; every lemma below then holds for those columns for the same reason as in its case (a), so no padding of the ground set is needed.

Definition 2 (the trajectory vector, including the case $b_t \in O$). *For $t = 0, \dots, K-1$ put $d_t = d_{b_t}(S^t)$, and for $t = 0, \dots, K$ and $i = 1, \dots, K$ put*

$$g_{t,i} = \begin{cases} d_{o_i}(S^t), & o_i \notin S^t, \\ 0, & o_i \in S^t. \end{cases} \quad (3)$$

The second line of equation 3 is the zeroing convention. It is the device that handles the case in which greedy selects an element of O : as soon as $b_t = o_i$, the variable $g_{t+1,i}$ and all later ones are set to zero rather than left undefined. Every lemma below is stated and proved with this convention in force, and in each proof the case $b_t = o_i$ appears as a separate case (c). Note that the convention is consistent with reading $g_{t,i}$ as a marginal gain, because an element already in S^t contributes no further gain to it.

Also set $r_t = 1 - \sum_{s < t} d_s$, the residual after t steps. Because $\sum_{s < t} d_s$ telescopes to $f(S^t) - f(\emptyset) = f(S^t)$, we have

$$r_t = f(O) - f(S^t). \quad (4)$$

The reduced LP of `code/reduced_lp.py` is, in these variables,

$$\begin{aligned} \min \quad & \sum_{t=0}^{K-1} d_t \\ \text{s.t.} \quad & \sum_{i=1}^K g_{t,i} \geq r_t & (\text{sum}(t)), \quad t = 0, \dots, K-1, \\ & d_t \geq g_{t,i}/\eta & (\text{pred}(t, i)), \\ & g_{t+1,i} \leq g_{t,i} & (\text{mono}(t, i)), \\ & (1 - 1/\eta) g_{t+1,i} \geq g_{t,i} - d_t & (\text{cons}(t, i)), \\ & d_t \geq 0, \quad g_{t,i} \geq 0. \end{aligned} \quad (5)$$

1.2 THE COHERENCE LEMMA

The fourth family rests on the coherence lemma (RESEARCH_STATE R3, called the consistency lemma in earlier drafts). We restate it and give its proof, so that the present section is self contained.

Lemma 3 (coherence, R3). *Let $S \subseteq N$ and let $e, e' \notin S$ with $e \neq e'$ satisfy $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$. Then*

$$d_e(S \cup \{e'\}) \geq d_{e'}(S \cup \{e'\})/\eta, \quad (6)$$

$$(1 - 1/\eta) d_{e'}(S \cup \{e\}) \geq d_{e'}(S) - d_e(S). \quad (7)$$

Proof. Expanding $\tilde{f}(S \cup \{e, e'\}) - \tilde{f}(S)$ in the two possible orders gives the identity $\tilde{d}_e(S) + \tilde{d}_{e'}(S \cup \{e\}) = \tilde{d}_{e'}(S) + \tilde{d}_e(S \cup \{e'\})$. This step uses nothing beyond the definition of a set function.

Subtracting and using $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$ yields $\tilde{d}_{e'}(S \cup \{e\}) \leq \tilde{d}_e(S \cup \{e'\})$. This step uses the greedy choice, in the form of the hypothesis on S .

Now $d_{e'}(S \cup \{e\}) \leq \eta_u \tilde{d}_{e'}(S \cup \{e\})$; this step uses the lower band. Then $\eta_u \tilde{d}_{e'}(S \cup \{e\}) \leq \eta_u \tilde{d}_e(S \cup \{e'\})$ by the previous display, and $\eta_u \tilde{d}_e(S \cup \{e'\}) \leq \eta_u \eta_o d_e(S \cup \{e'\}) = \eta d_e(S \cup \{e'\})$; this step uses the upper band. Chaining the three gives equation 6.

For equation 7, the same two-order expansion applied to f gives $d_e(S) + d_{e'}(S \cup \{e\}) = d_{e'}(S) + d_e(S \cup \{e'\})$, hence $d_{e'}(S) - d_e(S) = d_{e'}(S \cup \{e\}) - d_e(S \cup \{e'\})$. This step uses nothing beyond the definition of a set function. Substituting equation 6 in the form $-d_e(S \cup \{e'\}) \leq -d_{e'}(S \cup \{e\})/\eta$ gives $d_{e'}(S) - d_e(S) \leq (1 - 1/\eta) d_{e'}(S \cup \{e\})$, which is equation 7. This step uses equation 6 together with $\eta \geq 1$, so that the factor $1 - 1/\eta$ is nonnegative. $\square$

1.3 THE FOUR FAMILIES ARE VALID

Throughout, $t \in \{0, \dots, K-1\}$ and $i \in \{1, \dots, K\}$, and the three cases are (a) $o_i \in S^t$; (b) $o_i \notin S^t$ and $o_i \neq b_t$; (c) $o_i = b_t$, which is the case $b_t \in O$.

Lemma 4 (sign constraints). *$d_t \geq 0$ and $g_{t,i} \geq 0$ for all t, i .*

Proof. Each of these numbers is either a marginal gain of f or, under the zeroing convention, equal to 0. This step uses monotonicity of f . $\square$

Lemma 5 (sum(t): coverage). *For every t , $\sum_{i=1}^K g_{t,i} \geq r_t$.*

Proof. Let $I_t = \{i : o_i \notin S^t\}$, so that by the zeroing convention $\sum_{i=1}^K g_{t,i} = \sum_{i \in I_t} d_{o_i}(S^t)$. This step uses Definition 2, and it is exactly where the case $b_s \in O$ for some $s < t$ is absorbed: the indices of the elements of O already picked drop out of the sum, and they drop out on both sides of the argument below.

Order $I_t = \{i_1, \dots, i_m\}$ and telescope

$$f(S^t \cup O) - f(S^t) = \sum_{j=1}^m \left[f(S^t \cup \{o_{i_1}, \dots, o_{i_j}\}) - f(S^t \cup \{o_{i_1}, \dots, o_{i_{j-1}}\}) \right] \leq \sum_{j=1}^m d_{o_{i_j}}(S^t).$$

This step uses submodularity, applied once per term: the gain of o_{i_j} at the larger set $S^t \cup \{o_{i_1}, \dots, o_{i_{j-1}}\}$ is at most its gain at S^t .

Next, $f(S^t \cup O) \geq f(O)$. This step uses monotonicity.

Combining the two displays with equation 4,

$$\sum_{i=1}^K g_{t,i} \geq f(S^t \cup O) - f(S^t) \geq f(O) - f(S^t) = r_t. \quad \square$$

Lemma 6 (pred(t, i): greedy choice inside the band). *For every t, i , $d_t \geq g_{t,i}/\eta$.*

Proof. Case (a): $g_{t,i} = 0$ and $d_t \geq 0$, so the inequality holds. This step uses monotonicity (through Lemma 4).

Case (b): chain

$$d_t = d_{b_t}(S^t) \geq \frac{\tilde{d}_{b_t}(S^t)}{\eta_o} \geq \frac{\tilde{d}_{o_i}(S^t)}{\eta_o} \geq \frac{d_{o_i}(S^t)}{\eta_u \eta_o} = \frac{g_{t,i}}{\eta}.$$

The first inequality uses the upper band at (S^t, b_t) , rearranged as $d \geq \tilde{d}/\eta_o$. The second inequality uses the greedy choice, since $o_i \notin S^t$ makes o_i an eligible candidate at step t . The third inequality uses the lower band at (S^t, o_i) , rearranged as $\tilde{d} \geq d/\eta_u$.

Case (c): here $g_{t,i} = d_{o_i}(S^t) = d_{b_t}(S^t) = d_t$, and the claim reads $d_t \geq d_t/\eta$, which holds because $d_t \geq 0$ and $\eta \geq 1$. This step uses monotonicity and $\eta \geq 1$. Note that no band inequality is needed in this case, so the constraint does not degrade when greedy happens to pick an optimal element. $\square$

Lemma 7 (mono(t, i): single-element submodularity). *For every t, i , $g_{t+1,i} \leq g_{t,i}$.*

Proof. Case (a): $o_i \in S^t \subseteq S^{t+1}$, so both sides are 0. This step uses the zeroing convention only.

Case (b): $o_i \notin S^{t+1}$ as well, so $g_{t+1,i} = d_{o_i}(S^{t+1}) \leq d_{o_i}(S^t) = g_{t,i}$ because $S^t \subseteq S^{t+1}$. This step uses submodularity, in its diminishing-returns form for a single element.

Case (c): $o_i = b_t \in S^{t+1}$, so $g_{t+1,i} = 0$ while $g_{t,i} = d_t \geq 0$. This step uses monotonicity. This is the only place in the four families where the case $b_t \in O$ replaces a submodularity argument by a monotonicity argument. $\square$

Lemma 8 (cons(t, i): coherence). *For every t, i , $(1 - 1/\eta) g_{t+1,i} \geq g_{t,i} - d_t$.*

Proof. Case (a): the left side is 0 and the right side is $0 - d_t \leq 0$. This step uses monotonicity.

Case (b): apply Lemma 3 with $S = S^t$, $e = b_t$ and $e' = o_i$. The hypothesis $\tilde{d}_{b_t}(S^t) \geq \tilde{d}_{o_i}(S^t)$ of that lemma holds because o_i is eligible at step t ; this step uses the greedy choice. Inequality equation 7 then reads

$$(1 - 1/\eta) d_{o_i}(S^t \cup \{b_t\}) \geq d_{o_i}(S^t) - d_{b_t}(S^t),$$

which is the claim, since $S^{t+1} = S^t \cup \{b_t\}$ and $o_i \notin S^{t+1}$ give $d_{o_i}(S^{t+1}) = g_{t+1,i}$. This step uses R3(ii).

Case (c): $g_{t+1,i} = 0$ by the zeroing convention and $g_{t,i} - d_t = d_{b_t}(S^t) - d_{b_t}(S^t) = 0$, so the constraint holds with equality. This step uses Definition 2 only. In particular Lemma 3, which requires $e \neq e'$, is never invoked in this case, which is what closes the gap flagged in R6. $\square$

1.4 CONSEQUENCE

Lemma 9 (relaxation). *For every instance $(f, \tilde{f})$ satisfying Definition 1 with $\eta = \eta_u \eta_o$, every optimal set O and every run of predictive greedy, the vector $(d_t, g_{t,i})$ of Definition 2 is feasible for equation 5 and its objective value equals $f(S^K)/f(O)$. Hence the optimum of equation 5 is a lower bound on the worst-case ratio of predictive greedy at budget K and error η .*

Proof. Feasibility is Lemmas 4 to 8. The objective is $\sum_{t < K} d_t = f(S^K)$ by telescoping, and $f(O) = 1$ by the normalisation, so the objective equals the ratio. This step uses the definition of d_t and the normalisation. Since every realisable run gives a feasible point, the minimum over the feasible set is at most the minimum over realisable runs, that is, at most the worst-case ratio. $\square$

Remark 10 (what is and is not claimed). *Lemma 9 gives one direction only. The reverse direction, that every feasible point of equation 5 is realised by some instance, is not claimed here; it has been checked against the full-lattice LP for $K \leq 5$ and at nineteen (K, η) test points. The status of the material in this section is [HAND-PROOF-UNREVIEWED]: no oracle certifies these derivations.*

Remark 11 (which rulers the four families tolerate). *Families $\text{sum}(t)$ and $\text{mono}(t, i)$ use no error notion at all. Family $\text{pred}(t, i)$ uses the band only at the greedy state S^t , so it stays valid when η is replaced by the trajectory error η^{tr} , and, directly from its definition as a ratio of true gains at greedy states, when it is replaced by the selection error η^{sel} . Family $\text{cons}(t, i)$ invokes Lemma 3, whose proof also uses the band at the state $S^t \cup \{o_i\}$, which is not on the greedy trajectory; so cons is not covered by either restricted ruler without a further assumption. This is the reason the L_K bound, whose proof uses only sum and pred , is stated for η^{sel} , while the exact reduced-LP value is stated for the global η .*

Remark 12 (which hypothesis each family consumes). *Family $\text{sum}(t)$ uses submodularity and monotonicity of f and no property of $\tilde{f}$. Family $\text{pred}(t, i)$ uses the greedy choice and both bands, and no property of f beyond monotonicity. Family $\text{mono}(t, i)$ uses submodularity, and monotonicity in case (c). Family $\text{cons}(t, i)$ uses the greedy choice and both bands through Lemma 3, and monotonicity in cases (a) and (c). No family uses submodularity of $\tilde{f}$, which is consistent with the predictor being an arbitrary set function.*

REFERENCES

Geoffrey E. Hinton, Simon Osindero, and Yee Whye Teh. A fast learning algorithm for deep belief nets. *Neural Computation*, 18:1527–1554, 2006.